

Multiple Choice (10 marks)

1. In Python 3, what is the output of `print tuple[0]` if `tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )`?
 - (a) ('abcd', 786 , 2.23, 'john', 70.2)
 - (b) abcd
 - (c) Error
 - (d) None of the above.
2. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
3. What is the value of this python expression: `4 + 3 % 2`?
 - (a) 1
 - (b) 6
 - (c) 4
 - (d) 5
4. Let `l = [1,2,3,4]`. What is `max(l)` in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. The boolean expression `a[i] == max || !(max != a[i])` can be simplified to
 - (a) `a[i] == max`
 - (b) `a[i] != max`
 - (c) `a[i] < max || a[i] > max`
 - (d) FALSE

True / False (10 marks)

Answer True or False

6. True or False: "abc" reversed in Python 3 results in "cba".
7. True or False: In Python 3, using the set() function on a list removes duplicates, resulting in {1, 2, 3, 4} from the list [1, 2, 2, 3, 4].
8. True or False: In Python, the expression 4 * 1 ** 4 evaluates to 4.
9. True or False: In Python 3, the '%' operator is used for floor division.
10. True or False: In Python 3, the expression 'abc'[-1] results in an error due to invalid indexing.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the RGB color model represents colors using values for red, green, and blue. How many bits are required for this representation, and why is this significant in digital imaging?
12. Discuss how providing laptops or tablets to all students can help bridge the digital divide in education. Analyze the potential benefits and challenges of such a policy.

End of Paper